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System and method for identifying subsurface structures

Abstract

A subsurface structure identification system includes one or more processors and a memory coupled to the one or more processors. The memory is encoded with instructions that when executed by the one or more processors cause the one or more processors to provide a convolutional neural network trained to identify a subsurface structure in an input migrated seismic volume, and to partition the input migrated seismic volume into multi-dimensional sub-volumes of seismic data. The instructions also cause the one or more processors to process each of the multi-dimensional sub-volumes of seismic data in the convolutional neural network, and identify the subsurface structure in the input migrated seismic volume based on a probability map of the input migrated seismic volume generated by the convolutional neural network.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage Entry of International Application No.

PCT/US2019/031687, filed on May 10, 2019. The above application is incorporated herein by reference.

BACKGROUND

(2) Operations, such as surveying, drilling, wireline testing, completions, production, planning and field analysis, are typically performed to locate and gather valuable downhole fluids. Surveys are often performed using acquisition methodologies, such as seismic surveys to generate maps of underground formations. These formations are often analyzed to determine the presence of subterranean assets, such as valuable fluids or minerals, or to determine if the formations have characteristics suitable for storing fluids. Hydrocarbons, such as oil and natural gas are examples of such subterranean assets.

(3) Seismic surveying is generally performed by imparting energy to the earth at one or more

source locations, for example, by way of controlled explosion, mechanical input, etc. Return energy is then measured at surface receiver locations at varying distances and azimuths from the source location. The travel time of energy from source to receiver, via reflections and refractions from interfaces of subsurface strata, indicates the depth and orientation of such strata. Seismic data, as collected via the receiver, within a volume of interest is referred to as a seismic volume. A seismic volume can be displayed as seismic images based on different sampling resolutions and viewing orientations as well as subject to various different seismic amplitude processing techniques to enhance or highlight seismic reflection patterns.

(4) Seismic images indirectly show the distribution of material deposited over large areas. The spatial (and temporal) variability of stacking patterns, or sequences, observed in seismic images relates to depositional environments and post-depositional processes, such as erosion and tectonic activity. Seismic interpretation examines seismic data (e.g., location and time or depth) to identify subsurface structures that may be indicative of subterranean assets. During seismic interpretation, reflection patterns in the seismic images linking depositional environments and vertical stacking order to sequence of deposition and, thus, relative timing, enables the geological history of the subsurface to be deciphered and leads to estimation of probable sedimentary characteristics. In this manner, a potential hydrocarbon reservoir may be identified and analyzed based on interpretation and analysis of seismic reflection data.

SUMMARY

(5) A system and method for automated identification of subsurface structures by analyzing seismic data are disclosed herein. In one implementation, a subsurface structure identification system includes one or more processors and a memory coupled to the one or more processors. The memory is encoded with instructions that when executed by the one or more processors cause the one or more processors to provide a convolutional neural network trained to identify a subsurface structure in an input migrated seismic volume, and to partition the input migrated seismic volume into multi-dimensional sub-volumes of seismic data. The instructions also cause the one or more processors to process each of the multi-dimensional sub-volumes of seismic data in the convolutional neural network, and identify the subsurface structure in the input migrated seismic volume based on a probability map of the input migrated seismic volume generated by the convolutional neural network. The multi-dimensional sub-volumes are at least three dimensional.

(6) In another implementation, a method for identifying a subsurface structure includes applying a plurality of multi-dimensional sub-volumes of a first input migrated seismic volume to train a volumetric convolutional neural network to identify the subsurface structure in the input migrated seismic volume, and produce a trained convolutional neural network. A second input migrated seismic volume is partitioned into multi-dimensional sub-volumes of seismic data. Each of the multi-dimensional sub-volumes of the second input migrated seismic volume is processed by the trained convolutional neural network. A probability map for the second input migrated seismic volume is generated based on output of the trained convolutional neural network. The subsurface structure is identified in the second input migrated seismic volume based on the probability map. The multi-dimensional sub-volumes are at least three dimensional.

(7) In a further implementation, a non-transitory computer-readable medium encoded with instructions that when executed cause one or more processors to partition an input migrated seismic volume into multi-dimensional sub-volumes of seismic data, and apply the multi-dimensional sub-volumes to produce a trained convolutional neural network by training a volumetric convolutional neural network to identify a subsurface structure in the input migrated seismic volume. The multi-dimensional sub-volumes are at least three dimensional.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a detailed description of various examples, reference will now be made to the accompanying drawings in which:
- (2) FIG. 1 shows a block diagram for a system for identifying subsurface structures in a seismic volume in accordance with the present disclosure;
- (3) FIG. 2 shows a schematic level diagram for a volumetric convolutional neural network applied in the system of FIG. 1;
- (4) FIG. 3 shows an example of three-dimensional sub-volumes of seismic data provided to the volumetric convolutional neural network for use in identifying a subsurface structure;
- (5) FIG. 4 shows a flow diagram for an example method for identifying subsurface structures using a volumetric convolutional neural network in accordance with the present disclosure;
- (6) FIG. 5 shows a flow diagram for an example method for selecting sub-volumes of an input migrated seismic volume for use in training a volumetric convolutional neural network to identify subsurface structures in accordance with the present disclosure;
- (7) FIG. 6 shows a flow diagram for an example method for accelerated training of a volumetric convolutional neural network to identify subsurface structures in accordance with the present disclosure;
- (8) FIGS. 7 and 8 show flow diagrams for example methods for selecting a pixel state if overlapping sub-volumes of an input migrated seismic volume are processed by a volumetric convolutional neural network in accordance with the present disclosure;
- (9) FIG. 9 shows an example of a sub-volume of an input migrated seismic volume selected based on probability of sub-surface structure presence for use in training a volumetric convolutional neural network in accordance with the present disclosure;
- (10) FIG. 10 shows an example of construction of a binary mask for a segmented seismic volume based on non-overlapping sub-volumes in accordance with the present disclosure;
- (11) FIG. 11 shows an example of construction of a binary mask for a segmented seismic volume based on overlapping sub-volumes in accordance with the present disclosure;
- (12) FIG. 12 shows an example of a subsurface structure identified using a volumetric convolutional neural network in accordance with the present disclosure; and
- (13) FIG. 13 shows a block diagram for an example computing system suitable for implementation of a system for identifying subsurface structures in seismic volume in accordance with the present disclosure.

DETAILED DESCRIPTION

- (14) Certain terms have been used throughout this description and claims to refer to particular system components. As one skilled in the art will appreciate, different parties may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function. In this disclosure and claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection or through an indirect connection via other devices and connections. The recitation “based on” is intended to mean “based at least in part on.” Therefore, if X is based on Y, X may be a function of Y and any number of other factors. A “channel” is a linear, commonly concave-based depression through which water and sediment flow and into which sediment can be deposited in distinctive, often elongated bodies.
- (15) Interpretation of large seismic volumes is a daunting task, especially so if performed manually. The amount of seismic data collected, and available for interpretation, is expected to increase as the need for energy increases, and seismic data acquisition techniques improve to reduce the cost of seismic surveys. Accordingly, systems that automate the process of seismic interpretation are

needed to reduce the overall time and expense associated with seismic interpretation.

(16) The seismic interpretation systems and methods disclosed herein reduce the time required to identify subsurface geological structures by using machine learning techniques to automate geo-body segmentation. A geo-body is a volume of rock material bounded by a surface that delineates a specific object with measurable geometry. Conventional seismic data interpretation can be time consuming and extremely expensive. Systems of the present disclosure may reduce the time required for geo-body segmentation by a factor of 1000 or more, with attendant reduction in the cost of performing the segmentation.

(17) The seismic interpretation system includes a convolutional neural network trained to identify geo-bodies, such as channel geo-bodies or salt geo-bodies. The convolutional neural network receives an input migrated seismic volume as input and processes the input migrated seismic volume to identify the geo-bodies in accordance with the training applied to the convolutional neural network. As used herein, an “input migrated seismic volume” refers to regularly sampled subsurface images resulting from seismic data processing, such as a seismic depth migration result, a seismic time migration result, subsequent seismic time or depth inversion results, but also cruder binned and stacks. Such “images” are typically in three dimensions, but could be of higher dimension. In some implementations of the seismic interpretation system, the convolutional neural network is a fully convolutional network that processes a two-dimensional input migrated seismic volume to produce two-dimensional slices of a geo-body. In such systems, the two-dimensional slices are combined to produce a three-dimensional image of the geo-body. In other implementations of the seismic interpretation system, the convolutional neural network is a volumetric convolutional neural network that processes a three-dimensional input migrated seismic volume to produce a three-dimensional geo-body volume.

(18) The training of the convolutional neural networks includes generating a set of training data with geo-bodies labeled by one or more seismic data interpretation experts. For a fully convolutional network, the sets of training data include two-dimensional input migrated seismic volumes, and for a volumetric convolutional neural network, the sets of training data include three-dimensional input migrated seismic volumes. The training data is applied to the convolutional neural network, the results are compared to the labeling, and the weight values applied in the convolutional neural network are adjusted with each pass to improve the results produced in segmenting the training data. In some implementations of the seismic interpretation system, the training process is accelerated by simultaneous launching multiple instances of a convolutional neural network and training the different convolutional neural networks using different hyperparameters. The training results are scored and an instance of the trained convolutional neural network producing the best segmentation score may be selected for deployment. In this manner, the convolutional neural network training time and cost may be reduced (e.g., by a factor of 10 or more).

(19) FIG. 1 shows a block diagram for a system **100** for identifying subsurface structures in a seismic volume in accordance with the present disclosure. The system **100** includes seismic volume partitioning **104**, a volumetric convolutional neural network **106**, and image assembly **108**. The system **100** receives as input an input migrated seismic volume **102**. The input migrated seismic volume **102** may be provided to the system **100** in SEG-Y format or other format used to record and/or store seismic data.

(20) The input migrated seismic volume **102** may be relatively large (e.g., up to terabytes). The input migrated seismic volume **102** is too large to be processed by the volumetric convolutional neural network **106** as a unit. To facilitate processing by the volumetric convolutional neural network **106**, the seismic volume partitioning **104** subdivides or partitions the input migrated seismic volume **102** into multiple three-dimensional sub-volumes **112** of a size suitable for processing by the volumetric convolutional neural network **106**. For example, the seismic volume partitioning **104** may partition the input migrated seismic volume **102** into three-dimensional sub-

volumes **112** of $128 \times 128 \times 64$ pixels, or other sub-volume size suitable for processing by the volumetric convolutional neural network **106**. In various implementations of the system **100**, the seismic volume partitioning **104** partitions the input migrated seismic volume **102** into non-overlapping three-dimensional sub-volumes **112** or overlapping three-dimensional sub-volumes **112**. The seismic volume partitioning **104** may also normalize the amplitude of the input migrated seismic volume provided to the volumetric convolutional neural network **106** to bring the seismic amplitudes within a predetermined range.

(21) The volumetric convolutional neural network **106** processes the three-dimensional sub-volumes **112** to generate probability values **114** that represent the likelihood that each pixel of three-dimensional sub-volumes **112** is part of subsurface structure, such as a channel. The volumetric convolutional neural network **106** may include a plurality of volumetric convolutional layers that process the three-dimensional sub-volumes **112** to identify various features that are indicative of a subsurface structure, such as a channel, that the volumetric convolutional neural network **106** is trained to identify.

(22) The image assembly **108** receives the probability values **114** generated by the volumetric convolutional neural network **106** and combines the probability values **114** to generate the segmented volume **110**. The segmented volume **110** may be equivalent in size to the input migrated seismic volume **102**, and include information identifying sub-surface structures in the input migrated seismic volume **102**.

(23) FIG. 2 shows a schematic level diagram for a volumetric convolutional neural network **200**. The volumetric convolutional neural network **200** is an implementation of the volumetric convolutional neural network **106** of the system **100**. The volumetric convolutional neural network **200** processes the sub-volume **202** to identify sub-surface structures in the sub-volume **202**. The sub-volume **202** is an instance of the three-dimensional sub-volumes **112**. The volumetric convolutional neural network **200** includes a compression path **203** that reduces resolution and a decompression path **205** that increases resolution to the resolution of the sub-volume **202**. The compression path **203** includes stage **204**, stage **206**, stage **208**, stage **210**, and stage **212**. The decompression path **205** includes stage **214**, stage **216**, stage **218**, and stage **220**. Volumetric convolutions ($5 \times 5 \times 5$ voxels) are performed at each stage (each stage includes 1-3 convolutional layers) to extract features that are indicative of sub-surface structures from the sub-volume **202**. For example, stage **204** and stage **206** may identify fine low-level features, such as dots, lines, curves, and high intensity pixels in the sub-volume **202**. Stage **208** and stage **210** may identify mid-level features, such as corners, blobs, and contours in the sub-volume **202**. Stage **212** may identify high-level features such as broad patterns and object locations. Stage **214**, stage **216**, stage **218**, and stage **220** combine the low-level features and the high-level features identified in the compression path **203**.

(24) In each stage, input is provided to the convolution layers for feature identification and added to the output of the stage. In the compression path **203**, each successive stage includes more features identification channels (e.g., twice the number of channels in each successive stage). In the decompression path **205**, each successive stage includes fewer features identification channels (e.g., half the number of channels in each successive stage). Following the stage **204**, the stage **206**, the stage **208**, and the stage **210** down-conversion convolution is provided to reduce resolution. Following the stage **212**, the stage **214**, the stage **216**, and the stage **218** up-conversion deconvolution is provided to increase resolution. Features extracted in the compression path **203** are forwarded to the decompression path **205** to reduce loss of detail in the compression path **203**. For example, output of the stage **204** is passed to the input of the stage **220**, output of the stage **206** is passed to the input of the stage **218**, etc. Parametric rectified linear unit (PreLu) layers are included with each convolution.

(25) Implementations of the volumetric convolutional neural network **200** may apply a weighted loss function, such as a weighted binary cross entropy loss function. Values of the weighted loss

function are derived from a seismic volume (e.g., the fraction of the seismic volume that is a sub-surface structure).

(26) The stage **220** outputs two feature maps **222**. Soft-max processing **224** may be applied to the **222** to produce a segmentation of the sub-volume **202** that identifies pixels of the sub-volume **202** as either belonging to a sub-surface structure or background.

(27) FIG. **3** shows an example of three-dimensional sub-volumes **302-306** of an input migrated seismic volume provided to the volumetric convolutional neural network **200** for use in identifying a subsurface structure. The three-dimensional sub-volumes **302-306** are examples of the three-dimensional sub-volumes **112** and the three-dimensional sub-volume **202**.

(28) FIG. **4** shows a flow diagram for an example method **400** for identifying subsurface structures using a volumetric convolutional neural network in accordance with the present disclosure. Though depicted sequentially as a matter of convenience, at least some of the actions shown can be performed in a different order and/or performed in parallel. Additionally, some implementations may perform only some of the actions shown.

(29) In block **402**, the volumetric convolutional neural network **106** is trained to identify a sub-surface structure in the input migrated seismic volume **102**. The training may include acquiring seismic data that includes images of one or more sub-surface structures. The seismic data is analyzed to identify the sub-surface structures, and identified sub-surface structures are labeled. The labeled seismic data is partitioned to produce sub-volumes suitable for processing by the volumetric convolutional neural network **106**, and applied to train the volumetric convolutional neural network **106**. The training includes comparing results of processing by the volumetric convolutional neural network **106** to the labeled seismic data and back-propagating the results to the volumetric convolutional neural network **106** to improve identification of the various features of the subsurface structure.

(30) In block **404**, training of the volumetric convolutional neural network **106** is complete, and the trained volumetric convolutional neural network **106** is to be applied to process the input migrated seismic volume **102**. The seismic volume partitioning **104** subdivides the input migrated seismic volume **102** into three-dimensional sub-volumes **112** suitable for processing by the volumetric convolutional neural network **106**.

(31) In block **406**, the volumetric convolutional neural network **106** processes each of the three-dimensional sub-volumes **112** to identify a sub-surface structure in the three-dimensional sub-volumes **112**.

(32) In block **408**, probability values output by the volumetric convolutional neural network **106** for each of the three-dimensional sub-volumes **112** are combined by the image assembly **108** to generate a probability map for the input migrated seismic volume **102**.

(33) In block **410**, the probability values of the map produced in block **408** are resolved to identify a sub-surface structure in the input migrated seismic volume **102**. For example, pixels having a sub-surface structure probability value higher than a predetermined threshold are deemed to identify a sub-surface structure.

(34) In block **412**, an image of the sub-surface structure is generated. The image may label each pixel of the input migrated seismic volume **102** as a sub-surface structure identified by the volumetric convolutional neural network **106** or as background based on the results of processing the input migrated seismic volume **102** in blocks **404-410**.

(35) FIG. **5** shows a flow diagram for an example method **500** for selecting sub-volumes of an input migrated seismic volume for use in training a volumetric convolutional neural network to identify subsurface structures in accordance with the present disclosure. Though depicted sequentially as a matter of convenience, at least some of the actions shown can be performed in a different order and/or performed in parallel. Additionally, some implementations may perform only some of the actions shown. The method **500** may be performed as part of the operations of block **402** of the method **400**.

(36) Because sub-surface structures, such as channels, are extremely sparse in a seismic data volume (e.g., 1% of the pixels of seismic data volume may represent a subsurface structure to be identified), data applied to train the volumetric convolutional neural network **106** may be highly imbalanced in favor of background. To improve training of the volumetric convolutional neural network **106**, the method **600** applies a biased sampling technique to select the sub-volumes applied to train the volumetric convolutional neural network **106**.

(37) In block **502**, a probability value (p) is selected. The probability value will be applied to select sub-volumes of seismic data used to train the volumetric convolutional neural network **106**. The probability value may be selected at the start of training to optimize training of the volumetric convolutional neural network **106**.

(38) In block **504**, the probability value is used to select a pixel of the input migrated seismic volume. The pixel is selected to have probability (p) of being part of a subsurface structure.

(39) In block **506**, pixels about the selected pixel are chosen to form a sub-volume used to train the volumetric convolutional neural network **106**. For example, the selected pixel may be at the center of a cuboid arrangement of pixels forming a sub-volume of the seismic data volume. FIG. **9** shows an example of a sub-volume **904** of an input migrated seismic volume **900** selected based on pixel **902** having at least a probability (p) of being part of a sub-surface structure.

(40) FIG. **6** shows a flow diagram for an example method **600** for accelerated training of a volumetric convolutional neural network to identify subsurface structures in accordance with the present disclosure. Though depicted sequentially as a matter of convenience, at least some of the actions shown can be performed in a different order and/or performed in parallel. Additionally, some implementations may perform only some of the actions shown. The method **600** may be performed as part of the operations of block **402** of the method **400**. The method **600** may accelerate training of the volumetric convolutional neural network **106** by a factor of 10 or more.

(41) In block **602**, multiple instances of the volumetric convolutional neural network **106** are simultaneously trained using different hyperparameter values. The hyperparameters varied across the different instances of the volumetric convolutional neural network **106** may include: the probability applied to biased sampling of training sub-volumes, learning rate, batch size, training size, number of epochs, dropout probability, learning rate decay rate, etc. The instances of the volumetric convolutional neural network **106** may be implemented using virtual machines or other virtual computing devices provided via a cloud based computing system to allow for training of any number of instances of the volumetric convolutional neural network **106**.

(42) In block **604**, training of the multiple instances of the volumetric convolutional neural network **106** is complete, and the computing resources applied in the training are released. The results of the training are scored. For example, for each instance of the volumetric convolutional neural network **106**, an F1 score may be computed using the labeled training data and segmentation results generated by operation of the volumetric convolutional neural network **106**.

(43) In block **606**, the instance of the volumetric convolutional neural network **106** producing the highest score may be selected to be the trained volumetric convolutional neural network **106**.

(44) FIG. **7** shows a flow diagram for an example method **700** for selecting a pixel state if overlapping sub-volumes of an input migrated seismic volume are processed by the volumetric convolutional neural network **106**. Though depicted sequentially as a matter of convenience, at least some of the actions shown can be performed in a different order and/or performed in parallel. Additionally, some implementations may perform only some of the actions shown. The method **700** may be performed as part of the operations of block **408** of the method **400**.

(45) In the method **700**, the three-dimensional sub-volumes **112** provided to the volumetric convolutional neural network **106** are overlapping. Due to the overlapping three-dimensional sub-volumes **112**, for a given pixel of the input migrated seismic volume **102**, multiple probability values may be generated (e.g., a probability value for each three-dimensional sub-volume **112** that includes the pixel). In block **702**, the image assembly **108** statistically aggregates the probability

values generated for a given pixel. Statistical aggregation may include computation of arithmetic mean, weighted mean, median, harmonic mean, etc. of the probability values generated for the given pixel.

(46) In block **704**, the image assembly **108** applies a decision function to the probability produced by statistical aggregation in block **702** to determine pixel state. For example, if the probability exceeds a threshold, then the pixel is deemed part of a sub-surface structure, and is otherwise deemed part of background.

(47) FIG. **8** shows a flow diagram for an example method **800** for selecting a pixel state if overlapping sub-volumes of an input migrated seismic volume are processed by the volumetric convolutional neural network **106**. Though depicted sequentially as a matter of convenience, at least some of the actions shown can be performed in a different order and/or performed in parallel. Additionally, some implementations may perform only some of the actions shown. The method **800** may be performed as part of the operations of block **408** of the method **400**.

(48) In the method **800**, the three-dimensional sub-volumes **112** provided to the volumetric convolutional neural network **106** are overlapping. Due to the overlapping three-dimensional sub-volumes **112**, for a given pixel of the input migrated seismic volume **102**, multiple probability values may be generated (e.g., a probability value for each three-dimensional sub-volume **112** that includes the pixel). In block **802**, the image assembly **108** applies a decision function to each of the probability values generated for a given pixel. For example, the image assembly **108** may compare each of the probability values generated for the given pixel to a threshold value.

(49) In block **804**, the image assembly **108** sets pixel state based on majority of vote of the comparison results produced in block **802**. For example, if a majority of the probability values exceeds the threshold, then the pixel is deemed part of a sub-surface structure, and otherwise is deemed part of background.

(50) FIG. **10** shows an example of construction of a binary mask for a segmented seismic volume based on non-overlapping three-dimensional sub-volumes **112** in accordance with the present disclosure. The seismic volume partitioning **104** divides the input migrated seismic volume **1000** into non-overlapping sub-volumes **1002**, **1004**, **1006**, etc. While depicted as two-dimensional, the input migrated seismic volume **1000** and the sub-volumes **1002-1006** are three-dimensional. The sub-volumes **1002-1006** are provided to the volumetric convolutional neural network **106**, and the probability values **114** and/or the pixel state values generated by the volumetric convolutional neural network **106** for each sub-volume are likewise arranged in non-overlapping fashion to form the binary mask **1008**. The binary mask **1008** specifies whether each pixel of the input migrated seismic volume **1000** corresponds to a sub-surface structure or background.

(51) FIG. **11** shows an example of construction of a binary mask for a segmented seismic volume based on overlapping sub-volumes in accordance with the present disclosure. The seismic volume partitioning **104** divides the input migrated seismic volume **1100** into overlapping sub-volumes **1102**, **1104**, **1106**, etc. While depicted as two-dimensional, the input migrated seismic volume **1100** and the sub-volumes **1102-1106** are three-dimensional. The amount (the number of pixels) of overlap of the sub-volumes in X, Y, and Z directions can vary in implementations of the system **100**. The sub-volumes **1102-1106** are provided to the volumetric convolutional neural network **106**, and the probability values **114** are processed using the method **700** or the method **800** to determine the pixel state values for the binary mask **1108**. The binary mask **1108** specifies whether each pixel of the input migrated seismic volume **1100** corresponds to a sub-surface structure or background.

(52) FIG. **12** shows an example of a subsurface structure identified using the volumetric convolutional neural network **106**. In FIG. **12**, the area **1202** is area of sub-surface structure identified by a human expert. An input migrated seismic volume, with the area **1202** labeled as sub-surface structure was used to train the volumetric convolutional neural network **106**. After training, the input migrated seismic volume was processed by the system **100**, and the system **100** identified the area **1204** as sub-surface structure. The area **1204** closely correlates with area of the subsurface

structure identified by the human expert over which the area **1204** is overlaid in FIG. **12**.

(53) FIG. **13** shows a block diagram for a computing system **1300** suitable for implementation of the system **100**. Examples of the computing system **1300** may be applied to implement the seismic volume partitioning **104**, the volumetric convolutional neural network **106**, and/or the image assembly **108**. The computing system **1300** includes one or more computing nodes **1302** and secondary storage **1316** that are communicatively coupled (e.g., via the network **1318**). One or more of the computing nodes **1302** and associated secondary storage **1316** may be applied to provide the functionality of seismic volume partitioning **104**, the volumetric convolutional neural network **106**, and/or the image assembly **108** described herein.

(54) Each computing node **1302** includes one or more processors **1304** coupled to memory **1306**, a network interface **1312**, and the I/O devices **1314**. In various embodiments, a computing node **1302** may be a uniprocessor system including one processor **1304**, or a multiprocessor system including several processors **1304** (e.g., two, four, eight, or another suitable number). Processors **1304** may be any suitable processor capable of executing instructions. For example, in various embodiments, processors **1304** may be general-purpose or embedded microprocessors, graphics processing units (GPUs), or digital signal processors (DSPs) implementing any of a variety of instruction set architectures (ISAs). In multiprocessor systems, each of the processors **1304** may commonly, but not necessarily, implement the same ISA.

(55) The memory **1306** may include a non-transitory, computer-readable storage medium configured to store program instructions **1308** and/or data **1310** accessible by processor(s) **1304**. The memory **1306** may be implemented using any suitable memory technology, such as static random-access memory (SRAM), synchronous dynamic RAM (SDRAM), nonvolatile/Flash-type memory, or any other type of memory. Program instructions **1308** and data **1310** implementing the functionality disclosed herein are stored within memory **1306**. For example, instructions **1308** may include instructions that when executed by processor(s) **1304** implement the seismic volume partitioning **104**, the volumetric convolutional neural network **106**, and/or the image assembly **108** disclosed herein.

(56) Secondary storage **1316** may include volatile or non-volatile storage and storage devices for storing information such as program instructions and/or data as described herein for implementing the input migrated seismic volume **102**, the seismic volume partitioning **104**, the volumetric convolutional neural network **106**, the image assembly **108**, and/or the segmented volume **110**. The secondary storage **1316** may include various types of computer-readable media accessible by the computing node **1302** via the network interface **1312**. A computer-readable medium may include storage media or memory media such as semiconductor storage, magnetic or optical media, e.g., disk or CD/DVD-ROM, or other storage technologies.

(57) The network interface **1312** includes circuitry configured to allow data to be exchanged between the computing node **1302** and/or other devices coupled to the network **1318**. For example, the network interface **1312** may be configured to allow data to be exchanged between a first instance of the computing system **1300** and a second instance of the computing system **1300**. The network interface **1312** may support communication via wired or wireless data networks.

(58) The I/O devices **1314** allow the computing node **1302** to communicate with various input/output devices such as one or more display terminals, keyboards, keypads, touchpads, scanning devices, voice or optical recognition devices, or any other devices suitable for entering or retrieving data by one or more computing nodes **1302**. Multiple input/output devices may be present in a computing system **1300**.

(59) Those skilled in the art will appreciate that the computing system **1300** is merely illustrative and is not intended to limit the scope of embodiments. In particular, the computing system **1300** may include any combination of hardware or software that can perform the functions disclosed herein. Computing node **1302** may also be connected to other devices that are not illustrated, in some embodiments. In addition, the functionality provided by the illustrated components may in

some embodiments be combined in fewer components or distributed in additional components. Similarly, in some embodiments the functionality of some of the illustrated components may not be provided and/or other additional functionality may be available.

(60) The above discussion is meant to be illustrative of the principles and various embodiments of the present invention. Numerous variations and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.

Claims

1. A subsurface structure identification system, comprising: one or more processors; and a memory coupled to the one or more processors, and encoded with instructions that when executed by the one or more processors cause the one or more processors to: provide a convolutional neural network trained to identify subsurface structures in input migrated seismic volumes; partition a first input migrated seismic volume into three-dimensional sub-volumes, wherein the three-dimensional sub-volumes are overlapping portions of the first input migrated seismic volume; process each of the three-dimensional sub-volumes in the convolutional neural network to produce a respective plurality of feature maps associated with each three-dimensional sub-volume; generate, for each three-dimensional sub-volume, a respective segmentation that identifies portions of the three-dimensional sub-volume as either belonging to sub-surface structure or background based on the respective plurality of feature maps associated with the three-dimensional sub-volume; and identify a first subsurface structure in the first input migrated seismic volume based on a probability map of the first input migrated seismic volume generated based on the segmentations associated with the three-dimensional sub-volumes.
2. The subsurface structure identification system of claim 1, wherein the convolutional neural network comprises a convolutional layer configured to provide volumetric convolution of seismic data.
3. The subsurface structure identification system of claim 1, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to normalize amplitude of the first input migrated seismic volume.
4. The subsurface structure identification system of claim 1, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to statistically aggregate a plurality of probability values produced by processing the three-dimensional sub-volumes.
5. The subsurface structure identification system of claim 1, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to: apply a decision function to a plurality of probability values produced by processing the three-dimensional sub-volumes; and determine a pixel state based on a result of the decision function for each of the plurality of probability values.
6. The subsurface structure identification system of claim 1, wherein the first subsurface structure is a channel.
7. The subsurface structure identification system of claim 1, wherein a first three-dimensional sub-volume overlaps a second three-dimensional sub-volume by no more than 30%.
8. The subsurface structure identification system of claim 1, wherein the respective plurality of feature maps associated with each three-dimensional sub-volume is respective two feature maps associated with each three-dimensional sub-volume.
9. The subsurface structure identification system of claim 1, wherein the portions of each three-dimensional sub-volume that are identified as either belonging to sub-surface structure or background are represented by pixels.
10. A method for identifying a subsurface structure, comprising: applying a plurality of multi-

dimensional sub-volumes of a first input migrated seismic volume to train a volumetric convolutional neural network to identify a subsurface structure in the first input migrated seismic volume, and produce a trained convolutional neural network; partitioning a second input migrated seismic volume into multi-dimensional sub-volumes of seismic data, wherein the multi-dimensional sub-volumes are overlapping portions of the second input migrated seismic volume; processing each of the multi-dimensional sub-volumes of the second input migrated seismic volume by the trained convolutional neural network to produce a respective plurality of feature maps associated with each multi-dimensional sub-volume of the second input migrated seismic volume; generating, for each sub-volume of the second input migrated seismic volume, a respective segmentation that identifies portions of the multi-dimensional sub-volume as either belonging to sub-surface structure or background based on the respective plurality of feature maps associated with the multi-dimensional sub-volume; generating a probability map for the second input migrated seismic volume based on the segmentations associated with the multi-dimensional sub-volumes of the second input migrated seismic volume; and identifying the subsurface structure in the second input migrated seismic volume based on the probability map; wherein the multi-dimensional sub-volumes are at least three dimensional.

11. The method of claim 10, further comprising applying biased sampling to select the multi-dimensional sub-volumes of the first input migrated seismic volume that are applied to train the volumetric convolutional neural network.

12. The method of claim 11, wherein applying the biased sampling comprises: selecting a first pixel of the first input migrated seismic volume, wherein the first pixel has at least a predetermined probability of being part of the subsurface structure; and selecting a set of pixels about the first pixel to form one of the multi-dimensional sub-volumes of the first input migrated seismic volume.

13. The method of claim 12, further comprising varying the predetermined probability in different instances of training the volumetric convolutional neural network.

14. The method of claim 10, wherein training the volumetric convolutional neural network comprises: applying one or more back-propagation based optimization loops to minimize a weighted loss function that is applied to the volumetric convolutional neural network; and deriving values of weights of the weighted loss function from the first input migrated seismic volume.

15. The method of claim 10, further comprising normalizing amplitude of the second input migrated seismic volume prior to processing of the multi-dimensional sub-volumes of the second input migrated seismic volume by the trained convolutional neural network.

16. The method of claim 10, further comprising: statistically aggregating a plurality of probability values produced by processing the multi-dimensional sub-volumes; and applying a decision function to a result of the statistically aggregating to determine a state of a pixel corresponding to the plurality of probability values.

17. The method of claim 10, further comprising: applying a decision function to a plurality of probability values produced by processing the multi-dimensional sub-volumes; and determining a state of a pixel corresponding to the plurality of probability values based on a result of the decision function for the plurality of probability values.

18. The method of claim 10, further comprising: simultaneously training a plurality of instances of the volumetric convolutional neural network using the first input migrated seismic volume and different hyperparameter values in each of the plurality of instances; after completion of the simultaneous training, comparing accuracy of identification of the subsurface structure by each of the plurality of instances; and selecting an instance of the plurality of instances that most accurately identifies the subsurface structure as the trained convolutional neural network.

19. The method of claim 10, wherein a first multi-dimensional sub-volume overlaps a second multi-dimensional sub-volume by no more than 30%.
